

Self-Adaptive AdamW-Guided Optimization: A Learning-Driven Metaheuristic for Solving Complex Real-World Engineering Problems

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Abstract

With the growing complexity of continuous optimization problems in strongly coupled and black-box environments, this study proposes a novel adaptive gradient-guided metaheuristic, referred to as Self-Adaptive AdamW-guided Optimization (SAWG). Without requiring explicit gradient information, SAWG constructs population-based pseudo-gradients and systematically integrates key AdamW mechanisms, including adaptive moment estimation, step-size regulation, and weight decay, to guide efficient population updates. A stagnation-aware adaptive control strategy is further introduced to alleviate premature convergence and dynamically balance exploration and exploitation. To evaluate the optimization performance of SAWG, experiments were conducted on the CEC2017 and CEC2020 benchmark suites and eight engineering optimization problems. SAWG was also compared with nine other typical and novel high-performance optimizers. Experimental results and statistical analysis show that SAWG achieved excellent optimization performance in most test tasks and maintained strong adaptability and competitiveness in various numerical optimization problems. Therefore, SAWG can be regarded as a high-performance optimizer, providing a novel and effective method for solving complex numerical optimization tasks.

Keywords: AdamW; Benchmark test suite; Engineering design optimization; Pseudo-gradients; Metaheuristic optimization.

1. Introduction

Many critical tasks arising in engineering design, intelligent manufacturing, and energy scheduling[1] can be abstracted as complex continuous or hybrid optimization problems[2,3]. These problems are commonly associated with substantial nonlinearity, strong inter-variable coupling, and black-box characteristics[4], which collectively constrain the applicability of traditional gradient-based optimizers[5]. Since metaheuristic algorithms do not require an explicit analytical form of the objective function or accessible gradient information[6], they have become an important methodological paradigm for tackling such problems[7]. However, the No Free Lunch theorem[8] implies that no universal optimizer can maintain consistent superiority across all classes of optimization tasks, thereby underscoring the necessity of constructing specialized search mechanisms in accordance with problem-specific structural characteristics.

Although metaheuristic algorithms have progressed considerably, three fundamental difficulties still constrain their performance[9] on high-dimensional constrained opti-

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mization problems with complex structures. First, when the search process relies excessively on the current best individual or local refinement operators, population diversity is often diminished rapidly, thereby increasing the risk of premature convergence[10,11]; Ren et al. showed that maintaining diversity provably benefits evolutionary multimodal optimization[12]. Second, the coordination between exploration and exploitation is commonly regulated by fixed empirical parameters rather than mechanisms that respond adaptively to the search state and problem structure, making late-stage stagnation a recurrent issue[13]. Guan et al. proposed a feedback-based adaptive crossover-rate mechanism in evolutionary computation. Third, when the updating process is driven predominantly by stochastic perturbations, the resulting search trajectory may exhibit pronounced oscillatory behavior in complex spaces characterized by substantial scale heterogeneity, which consequently undermines both convergence efficiency and overall stability[14,15].

Inspired by parameter optimization mechanisms in machine learning, Doerr and Zheng theoretically analyze multi-objective evolutionary algorithms on multimodal objectives and show that heavy-tailed mutation and stagnation-aware strategies can substantially improve search efficiency.[16]; Li et al. also demonstrated the effectiveness of restarting and local search[17]; This study proposes a metaheuristic framework, denoted as Self-Adaptive AdamW-guided Optimization (SAWG), that synthesizes a stagnation-aware control mechanism, AdamW-based updating principles, and a pseudo-gradient-driven search[18] strategy. Specifically, SAWG constructs pseudo-gradient signals to impose globally directional guidance on population evolution, while first-order momentum and second-order statistical estimation are jointly exploited to adaptively modulate the update magnitude and search direction. Furthermore, a decoupled weight decay[15] mechanism is incorporated to reinforce the stability of the search dynamics.

The principal contributions of this study are summarized as follows:

1. This study formulates the Self-Adaptive AdamW-guided Optimization (SAWG) algorithm under an AdamW-based updating framework. The proposed method systematically integrates momentum memory and second-order statistical estimation, and further characterizes the dynamic search behavior of SAWG through visualization-based analyses of population diversity and search trajectories. In doing so, it establishes a methodological framework for population-based search from the perspective of machine-learning-driven optimization.
2. SAWG is systematically evaluated on the CEC 2017 benchmark suite under multiple dimensional settings. The results demonstrate that SAWG achieves stronger overall performance than a range of classical and advanced baseline optimizers across different problem scales, while its statistical superiority is further supported by significance tests. Specifically, SAWG exhibits consistent advantages in Friedman ranking, Wilcoxon tests, and aggregated performance indicators.
3. To examine its practical applicability, SAWG is further deployed on eight classical engineering optimization problems. The corresponding results show that the proposed method delivers favorable solution quality together with stable performance across all considered engineering tasks, thereby substantiating its robustness and engineering relevance in complex real-world environments.

The remainder of this paper is organized as follows. Section 2 discusses the motivation underlying the proposed algorithm. Section 3 introduces the overall framework and core mechanisms of SAWG. Section 4 presents the experimental setup and the associated result analysis. Section 5 reports the outcomes on engineering applications. Section 6 concludes the paper and outlines possible directions.

2. Related Work

To articulate the design motivation of the proposed SAWG algorithm, this section contrasts gradient-driven optimization with metaheuristic search from the perspective of optimization paradigms. We first review the core concepts and prerequisites of gradient-based methods, followed by an analysis of the advantages and inherent limitations of metaheuristics in black-box scenarios. Finally, we summarize recent advances in directional search mechanisms under derivative-free[18,19] conditions, highlighting the lack of systematic integration of mature gradient techniques, which lays the methodological foundation for SAWG.

2.1. Gradient-based Optimization in Machine Learning

Gradient-based optimization serves as the computational foundation of machine learning, relying on the assumption that the objective function is differentiable or admits reliable gradient approximations. Under this framework, advanced optimizers introduce momentum to suppress high-frequency oscillations[14], adaptive learning rates to mitigate scale inconsistencies, and weight decay to enhance generalization; for instance, Loshchilov and Hutter proposed AdamW, Kunstner et al. analyzed why Adam can outperform gradient descent in language models, and Zhou et al. further studied the convergence and generalization behavior of AdamW[20,21]. However, the efficacy of these methods strictly depends on gradient availability, rendering them inapplicable to complex scenarios where analytical derivatives are inaccessible or discontinuous.

2.2. Metaheuristic Algorithms for Black-box Optimization

For non-differentiable or computationally expensive black-box problems, metaheuristic algorithms perform global exploration through heuristic rules[22] and stochastic perturbations. Requiring minimal mathematical assumptions, they exhibit strong adaptability in complex landscapes; Hüttenrauch and Neumann developed a robust black-box optimization framework, while Zhang et al. systematically benchmarked zeroth-order optimization as a gradient-free alternative[23]. Nevertheless, their heavy reliance on randomness often lacks explicit directional constraints, making convergence susceptible to oscillation or stagnation in high-dimensional spaces[24,25]. Furthermore, balancing exploration and exploitation typically relies on empirical parameters, hindering adaptive dynamic adjustments based on search states; related recent work on task-dependent optimizer generation and history-aware black-box optimization also reflects this limitation[26].

2.3. Bridging Gradient-based Learning and Metaheuristic Search

Although gradient-based learning and metaheuristic search are traditionally viewed as distinct branches, recent studies have attempted to incorporate directional or memory-based mechanisms into derivative-free frameworks. Existing efforts include gradient matching for offline black-box optimization[19] by Hoang et al., pseudo-zeroth-order directional construction by Yue et al., and reinforced in-context black-box optimization by Song et al., all of which improve search behavior to some extent[18,27,28]. However, these methods primarily focus on local or partial modifications and fail to systematically integrate mature strategies from gradient optimizers, such as momentum, second-order moment estimation, and decoupled regularization, from a dynamics perspective[20,21]. How to embed these mechanisms into a derivative-free framework to construct a stable and memory-guided search process remains a critical open problem that this study aims to address.

3. SAWG Methodology

Based on the above analysis, this section presents the methodological foundation and overall framework of the proposed Self-Adaptive AdamW-guided Optimization (SAWG). The core objective of SAWG is to construct a swarm-based search mechanism endowed with gradient-learning characteristics under gradient-free black-box optimization settings. To this end, SAWG integrates AdamW-style adaptive update mechanisms, dynamic parameter modulation strategies, and stagnation-aware restart schemes to achieve coordinated global–local regulation of the search process. As a result, the proposed framework enables more stable and efficient optimization behavior when addressing complex, high-dimensional, and non-convex problems.

3.1. Gradient Descent as the Algorithmic Core

Gradient Descent (GD) is one of the most fundamental optimization paradigms in continuous optimization and machine learning. Its central idea is to iteratively update parameters along the negative gradient direction of the objective function in the parameter space, thereby gradually approaching the optimal solution. The gradient characterizes the direction of steepest ascent at a given point, and moving in the opposite direction yields the fastest decrease of the objective value. For differentiable objective functions, this mechanism provides a clear, stable, and well-interpretable descent direction.

In modern optimization theory, gradient descent has evolved far beyond its basic form with a fixed learning rate. Tian et al. show that rethinking weight decay in foundation-model fine-tuning can improve robustness and training stability by clarifying how AdamW-style decay affects optimization dynamics[29]. By incorporating mechanisms such as momentum accumulation, adaptive scale adjustment, and weight decay, gradient-based methods have achieved significantly improved robustness and convergence performance in non-convex, noisy, and high-dimensional optimization scenarios. Optimizers exemplified by AdamW jointly model first-order and second-order statistical information to adaptively regulate both search directions and step sizes, and are widely regarded as a major milestone in the development of gradient-based optimization methods[20,20,21].

Consider a continuous optimization problem with a differentiable objective function:

$$\min_{\mathbf{x} \in \Omega} f(\mathbf{x}), \quad \Omega = \{\mathbf{x} \mid \mathbf{l} \leq \mathbf{x} \leq \mathbf{u}\} \quad (1)$$

where $f: \mathbb{R}^d \rightarrow \mathbb{R}$ denotes the objective function, $\mathbf{x} = [x_1, \dots, x_d]^\top$ is the decision vector, Ω is a box-constrained feasible region, and $\mathbf{l}, \mathbf{u} \in \mathbb{R}^d$ represent the lower and upper bounds, respectively.

If f is differentiable over Ω and its gradient is accessible, Gradient Descent (GD) constructs descent updates via a local first-order approximation. Specifically, for a given iterate $\mathbf{x}_t$, the first-order Taylor expansion is given by:

$$f(\mathbf{x}_t + \Delta \mathbf{x}) \approx f(\mathbf{x}_t) + \nabla f(\mathbf{x}_t)^\top \Delta \mathbf{x}. \quad (2)$$

where, t denotes the iteration index, $\mathbf{x}_t$ is the current solution, and $\Delta \mathbf{x}$ is the update increment. To achieve the maximum decrease of this approximation under the step-size constraint $\|\Delta \mathbf{x}\|_2 \leq \eta_t$, the following constrained subproblem is formulated:

$$\min_{\Delta \mathbf{x}} \nabla f(\mathbf{x}_t)^\top \Delta \mathbf{x} \quad \text{s.t.} \quad \|\Delta \mathbf{x}\|_2 \leq \eta_t. \quad (3)$$

By the Cauchy–Schwarz inequality,

$$\nabla f(\mathbf{x}_t)^\top \Delta \mathbf{x} \geq -\|\nabla f(\mathbf{x}_t)\|_2 \cdot \|\Delta \mathbf{x}\|_2, \quad (4)$$

with equality attained when $\Delta \mathbf{x} = -\eta_t \frac{\nabla f(\mathbf{x}_t)}{\|\nabla f(\mathbf{x}_t)\|_2}$. Consequently, the direction of steepest descent corresponds to $-\nabla f(\mathbf{x}_t)$, yielding the classical GD update

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta_t \nabla f(\mathbf{x}_t), \quad (5)$$

where $\eta_t > 0$ denotes the learning rate that controls the update magnitude along the descent direction.

Despite its simplicity, GD is often challenged by nonconvex landscapes, noisy gradients, and heterogeneous scaling across dimensions. To address these limitations, modern optimization methods introduce momentum accumulation and second-order moment estimation, leading to adaptive gradient optimizers such as Adam and AdamW. These methods estimate the first-order momentum and second-order statistics of the gradient sequence $\mathbf{g}_t$, enabling more robust update directions and coordinate-wise adaptive step-size adjustment. This principle underpins the design of SAWG: even in black-box optimization settings where true gradients are unavailable, AdamW-level update mechanisms can be effectively exploited by constructing suitable surrogate gradient sequences.

3.2. Initialization Strategy

Each search agent is denoted by x_{ij} , which represents the position component of the i -th individual in the j -th dimension of the search space. All position components are required to satisfy the box constraints

$$\mathbf{l} \leq x_{ij} \leq \mathbf{u} \quad \forall i = 1, 2, \dots, N, \quad \forall j = 1, 2, \dots, \text{Dim} \quad (6)$$

where N denotes the population size and Dim is the problem dimensionality. The vectors $\mathbf{l}$ and $\mathbf{u}$ correspond to the lower and upper bounds of the decision variables, respectively. Consistent with conventional metaheuristic frameworks, SAWG initializes a population of candidate solutions randomly within the prescribed bounds to explore the search space, as given by

$$x_{ij} = \text{rand}(N, \text{Dim}) \times (\mathbf{l} - \mathbf{u}) + \mathbf{l} \quad (7)$$

where $\text{rand}(N, \text{Dim})$ denotes an $N \times \text{Dim}$ matrix with entries uniformly sampled from the interval $[0, 1]$.

At the initialization stage, SAWG identifies the individual with the best fitness value as the elite solution. For minimization problems, the position of this individual provides an approximation of the current global optimum. In parallel, the AdamW momentum states for each individual are initialized as

$$\mathbf{m}_i^{(0)} = \mathbf{0}, \quad \mathbf{v}_i^{(0)} = \mathbf{0}. \quad (8)$$

This initialization strategy prevents any historical bias at the early search stage, ensuring that the momentum information is accumulated adaptively and exclusively driven by the subsequent optimization process.

3.3. Gradient-Enhanced Update Mechanism of SAWG

3.3.1. Pseudo-Gradient Construction

In engineering design optimization and CEC benchmark testing, objective functions are typically treated as black boxes, for which analytical expressions are unavailable or non-differentiable, making the true gradient $\nabla f(\mathbf{x})$ inaccessible. To introduce gradient-like directional guidance in such gradient-free settings, SAWG constructs a computable pseudo-gradient by leveraging population-level statistics and elite information. This design yields a learnable search direction that can be naturally integrated into the subsequent AdamW update framework.

At iteration t , the population is denoted by $\{\mathbf{x}_{i,t}\}_{i=1}^N$, where N is the population size, $\mathbf{x}_{i,t} \in \mathbb{R}^D$ represents the position of the i -th individual, and D is the problem dimensionality. The population mean is defined as

$$\bar{\mathbf{x}}_t = \frac{1}{N} \sum_{j=1}^N \mathbf{x}_{j,t}. \quad (9)$$

The elite index j_t^* and elite position $\mathbf{x}_t^*$ are given by

$$j_t^* = \arg \min_{j \in \{1, \dots, N\}} f(\mathbf{x}_{j,t}), \quad \mathbf{x}_t^* = \mathbf{x}_{j_t^*, t}. \quad (10)$$

For each individual, two fundamental guidance vectors are constructed to provide continuously adjustable directional information between exploration and exploitation. The population-mean-guided vector $\mathbf{P}_{i,t}^{(1)}$ emphasizes exploratory and aggregative behavior, whereas the elite-guided vector $\mathbf{P}_{i,t}^{(2)}$ promotes exploitation toward the best-so-far solution:

$$\mathbf{P}_{i,t}^{(1)} = \bar{\mathbf{x}}_t - \tilde{\mathbf{x}}_{i,t}, \quad \mathbf{P}_{i,t}^{(2)} = \mathbf{x}_t^* - \mathbf{x}_{i,t}. \quad (11)$$

where, $\tilde{\mathbf{x}}_{i,t}^{(k)}$ denotes the k -th intermediate line state constructed within the same generation ($k = 1, \dots, K$, with $K = \lfloor D/2 \rfloor$). Its recursive formulation is

$$\tilde{\mathbf{x}}_{i,t} = \begin{cases} \tilde{\mathbf{x}}_{i,t}^{(1)} = (\omega_t + \alpha_i) \odot \mathbf{x}_{i,t}, \\ \tilde{\mathbf{x}}_{i,t}^{(k)} = \mathbf{x}_{i,t} + \sin(2\pi t D) \odot \tilde{\mathbf{x}}_{i,t}^{(k-1)}, \quad k > 1. \end{cases} \quad (12)$$

where ω_t is a time-varying random amplitude scalar and α_i is an individual-dependent phase coefficient, defined as

$$\omega = u_t \left(\frac{t^2}{T^2} - \frac{2t}{T} + \frac{1}{2} \right), \quad u_t \sim \mathcal{U}(0, 1), \quad (13)$$

$$\alpha_i = \cos(2\pi(1 - u_i)), \quad u_i \sim \mathcal{U}(0, 1). \quad (14)$$

To smoothly transition from broad exploration to focused exploitation, SAWG employs a hybrid weighting strategy that couples iteration progress with individual ranking. Let

$$\rho_i = \frac{i-1}{N-1} \in [0, 1], \quad c_2(i, t) = \frac{t}{T} \rho_i, \quad c_1(i, t) = 1 - c_2(i, t) \quad (15)$$

then the final guidance vector is obtained as

$$\mathbf{P}_{i,t} = c_1(i,t)\mathbf{P}_{i,t}^{(1)} + c_2(i,t)\mathbf{P}_{i,t}^{(2)}. \quad (16)$$

Before being integrated into AdamW, the guidance vector is normalized to eliminate scale discrepancies and improve numerical stability:

$$\tilde{\mathbf{P}}_{i,t} = \frac{\mathbf{P}_{i,t}}{\|\mathbf{P}_{i,t}\| + \varepsilon}. \quad (17)$$

where, $\varepsilon > 0$ is a small constant, $\|\cdot\|$ is the Euclidean norm. Finally, the black-box pseudo-gradient is constructed as

$$\mathbf{g}_{i,t} = \mathbf{x}_{i,t} - C_t \odot \tilde{\mathbf{P}}_{i,t}, \quad C_t = 1 - r, \quad r \sim \mathcal{U}(0,1) \quad (18)$$

where $\odot$ denotes element-wise multiplication, and C_t is a stochastic interaction-strength scalar shared across all dimensions.

It should be emphasized that the pseudo-gradient in SAWG is not a true derivative. Instead, it is generated from the relationship between the current position and the constructed guidance direction, which jointly captures the global population trend and the elite-driven attraction. This design provides continuously adjustable directional information between global exploration and local exploitation. By incorporating normalization and stochastic scaling, the pseudo-gradient maintains directional consistency while avoiding excessive determinism, thereby mitigating premature convergence and offering a unified interface for subsequent momentum integration and adaptive updates.

3.3.2. AdamW-guided Gradient-Enhanced Update

After the pseudo-gradient $\mathbf{g}_{i,t}$ is obtained, SAWG adopts the AdamW momentum integration mechanism to smooth the driving signal and introduce scale adaptivity, thereby constructing a gradient-type update structure under black-box optimization settings. For each individual, first-order and second-order momentum terms are maintained and updated according to

$$\mathbf{m}_{i,t} = \beta_1 \mathbf{m}_{i,t-1} + (1 - \beta_1) \mathbf{g}_{i,t}, \quad (19)$$

$$\mathbf{v}_{i,t} = \beta_2 \mathbf{v}_{i,t-1} + (1 - \beta_2) \mathbf{g}_{i,t}^{\odot 2}. \quad (20)$$

where, $\beta_1, \beta_2 \in (0,1)$ are the momentum decay coefficients (set to $\beta_1 = 0.9$ and $\beta_2 = 0.999$), and $\mathbf{g}_{i,t}$ denotes the pseudo-gradient defined in Equation (20).

To correct the bias introduced during the early iterations, bias-corrected estimates are computed as

$$\hat{\mathbf{m}}_{i,t} = \frac{\mathbf{m}_{i,t}}{1 - \beta_1^t}, \quad \hat{\mathbf{v}}_{i,t} = \frac{\mathbf{v}_{i,t}}{1 - \beta_2^t}. \quad (21)$$

The resulting adaptive update takes the form

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta_t \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t + \varepsilon}}, \quad (22)$$

where ε is a small constant for numerical stability and η_t is the learning rate. Building upon this formulation, AdamW introduces decoupled weight decay[20] to prevent the regular-

ization term from being suppressed by second-order adaptive scaling. The corresponding AdamW-style update step is given by

$$\Delta \mathbf{x}_{i,t} = \eta_t \left(\frac{\hat{\mathbf{m}}_{i,t}}{\sqrt{\hat{\mathbf{v}}_{i,t} + \varepsilon}} + \lambda_t \mathbf{x}_{i,t} \right). \quad (23)$$

To achieve adaptive regulation of the search scale, SAWG applies a cosine annealing strategy to both the learning rate η_t and the weight decay coefficient λ_t . Specifically, λ_{t0} is initialized to 0.01, and a piecewise strategy with $\eta_t = \eta_{\max}$ is adopted when $t < 0.4T$:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos \frac{\pi t}{T} \right) \quad (24)$$

$$\lambda_t = \lambda_{t0} \cdot 0.5 \cdot \left(1 + \cos \left(\frac{\pi \cdot t}{T} \right) \right). \quad (25)$$

where, $\eta_{\max}$ and $\eta_{\min}$ denote the upper and lower bounds of the learning rate, respectively, and T is the maximum number of iterations. This design enables the algorithm to maintain large update steps in the early stages to enhance exploration, while gradually reducing the step size in later stages to improve solution precision. Finally, the guidance point $\mathbf{G}_{i,t}$, influenced by both the adaptive learning rate and the best-performing unit, is defined as

$$\mathbf{G}_{i,t} = \mathbf{x}_{i,t} - \Delta \mathbf{x}_{i,t}. \quad (26)$$

3.3.3. Dynamic gradient interaction system

After the guidance point $\mathbf{G}_{i,t}$ is obtained, SAWG further incorporates differential interactions to enhance information exchange within the population. Two indices $a_1, a_2 \in 1, \dots, N \setminus i$ are randomly selected with $a_1 \neq a_2$, and a sign-based interaction intensity is defined as

$$\zeta_{i,t} = \frac{f(\mathbf{x}_{a_1,t}) - f(\mathbf{x}_{i,t})}{|f(\mathbf{x}_{a_1,t}) - f(\mathbf{x}_{i,t})| + \varepsilon}. \quad (27)$$

Based on the guidance point $\mathbf{G}_{i,t}$, two complementary candidate generation schemes are constructed:

$$\begin{cases} \mathbf{y}_{i,t}^{(a)} = \tilde{\mathbf{x}}_{i,t}^{(k)} + \zeta_{i,t} \mathbf{a}_t \odot (\mathbf{G}_{i,t} - \mathbf{x}_{a_1,t}) - \mathbf{a}_t \odot (\tilde{\mathbf{x}}_{i,t}^{(k)} - \mathbf{x}_{a_2,t}), \\ \mathbf{y}_{i,t}^{(b)} = \mathbf{x}_{a_1,t} + \mathbf{a}_t \odot (\mathbf{G}_{i,t} - \mathbf{x}_{a_2,t}). \end{cases} \quad (28)$$

where $\mathbf{a}_t = (1 - \frac{t}{T}) \cdot \text{rand}(1, \text{dim})$ is a random perturbation vector modulated by the dynamic scaling factor $(1 - \frac{t}{T})$, yielding a range of $[0, 1 - \frac{t}{T}]$. The candidate $\mathbf{y}_{i,t}^{(a)}$ corresponds to a double-differential perturbation around the intermediate line state, whereas $\mathbf{y}_{i,t}^{(b)}$ represents an AdamW-guided differential update anchored at $\mathbf{x}_{a_1,t}$. These two schemes jointly provide complementary search patterns.

To dynamically combine the two candidates at the dimensional level, the following switching rule is applied for each dimension $j \in 1, \dots, D$:

$$x_{i,t}^{\text{new}}(j) = \begin{cases} y_{i,t}^{(b)}(j), & \frac{u_1(j)}{k} > u_2(j), \\ y_{i,t}^{(a)}(j), & \text{otherwise,} \end{cases} \quad u_1(j), u_2(j) \sim \mathcal{U}(0, 1). \quad (29)$$

As k increases, the probability of selecting $\mathbf{y}^{(b)}$ gradually decreases, resulting in a progressive transition from aggressive to more conservative updates. Finally, boundary

handling and greedy acceptance are employed. The newly generated solution is projected onto the feasible domain, and greedy replacement is performed if improvement is achieved:

$$\mathbf{x}_{i,t}^{\text{new}} \leftarrow \min(\max(\mathbf{x}_{i,t}^{\text{new}}, \mathbf{l}), \mathbf{u}), \quad (30)$$

3.3.4. Stagnation Detection and Adaptive Restart Mechanism

To mitigate premature convergence induced by prolonged stagnation, SAWG monitors the search dynamics through the relative improvement rate of the global best solution:

$$\text{RI}_t = \frac{f_{t-1}^* - f_t^*}{|f_{t-1}^*| + \varepsilon}, \quad (31)$$

where $f_t^* = \min_i f(\mathbf{x}_{i,t})$ denotes the best fitness value at iteration t . When RI_t remains below a predefined stagnation threshold τ for a consecutive number of iterations, a restart mechanism is activated. Specifically, a proportion γ of the worst-ranked individuals (set to $0.2N$ in practice) are selected to undergo an opposition-based restart:

$$\mathbf{x}_{i,t} \leftarrow \mathbf{l} + \mathbf{u} - \mathbf{x}_{i,t} + \delta \odot (\mathbf{u} - \mathbf{l}), \quad (32)$$

where δ is a perturbation vector (to be consistent with the implementation, $\delta \sim \mathcal{N}(\mathbf{0}, 0.1^2 \mathbf{I})$ can be used). Meanwhile, the corresponding momentum states are reset as

$$\mathbf{m}_{i,t} = \mathbf{0}, \quad \mathbf{v}_{i,t} = \mathbf{0}. \quad (33)$$

By eliminating the influence of historical momentum, this mechanism prevents negative transfer to newly explored regions, thereby restoring population diversity and improving the robustness of the search process.

3.3.5. Lévy Flight–Based Global Exploration Compensation

Although the dynamic interaction mechanism improves local exploitation and information exchange, premature convergence may still arise in complex multimodal black-box optimization problems. To compensate for insufficient global exploration, SAWG probabilistically activates Lévy long-range jumps at the end of each generation with probability p_t :

$$p_t = p_{\min} + (p_{\max} - p_{\min})p_0(t), \quad (34)$$

$$p_0(t) = \frac{1}{1 + \exp\left(-s\left(\frac{t}{T} - \mu\right)\right)}. \quad (35)$$

where, $p_{\min}$ and $p_{\max}$ denote the lower and upper bounds of the triggering probability, while s and μ control the inflection point and steepness of the transition curve. Once activated, a Lévy step vector $\mathbf{L}_{i,t} \sim \text{Levy}(\beta)$ is generated, and a candidate solution is constructed as

$$\mathbf{x}'_{i,t} = \mathbf{x}_t^* + \mathbf{L}_{i,t} \odot \xi_t \odot \left(\mathbf{x}_t^* - \frac{2t}{T} \mathbf{x}_{i,t}\right), \quad (36)$$

where ζ_t is a time-varying random coefficient and β is the Lévy distribution index. The Lévy random vector can be generated using the Mantegna method:

$$\sigma_u = \left[\frac{\Gamma(1+\beta) \sin(\pi\beta/2)}{\Gamma\left(\frac{1+\beta}{2}\right) \beta 2^{(\beta-1)/2}} \right]^{1/\beta}, \quad (37)$$

$$u \sim \mathcal{N}(0, \sigma_u^2)_{n \times m}, \quad v \sim \mathcal{N}(0, 1)_{n \times m}, \quad \text{Levy}(N, D, \beta) = \frac{u}{|v|^{1/\beta}}. \quad (38)$$

Boundary clipping and greedy acceptance are subsequently applied to $\mathbf{x}'_{i,t}$, which injects long-tailed jump behavior while preserving the overall convergence tendency. This mechanism significantly increases the probability of escaping local optima and strengthens the global exploration capability. The pseudo-code of the proposed SAWG algorithm are displayed in Algorithm 1.

Algorithm 1 Pseudo-code of SAWG.

Require: Population size N , maximum number of iterations T , lower bound lb , upper bound ub , problem dimension dim , objective function f_{obj}
Ensure: Best solution position p_{best} , best fitness value f_{best} , convergence curve cg

- 1: Initialize population X and first- and second-order moment vectors m and v
- 2: Evaluate fitness of each individual and identify the global best solution p_{best}
- 3: Initialize stagnation counter and convergence curve cg
- 4: **while** $t \leq T$ **do**
- 5: Update learning rate and weight decay using cosine annealing using Equations (24) and (25)
- 6: Generate dynamic control parameters using Equations (13) and (14)
- 7: **for** $i = 1$ to N **do**
- 8: **for** $k = 1$ to $dim/2$ **do**
- 9: Construct perturbed base position using Equation (12)
- 10: Randomly select two distinct reference individuals using Equation (28)
- 11: Compute fitness-based directional coefficient using Equation (27)
- 12: Generate population-guided and elite-guided directions using Equations (11) and (16)
- 13: Update position using AdamW-guided pseudo-gradient using Equation (23)
- 14: Generate candidate solutions and select via greedy rule using Equation (30)
- 15: **end for**
- 16: **end for**
- 17: Perform Lévy-flight-based long jump with a given probability using Equation (36)
- 18: Update stagnation counter based on relative improvement
- 19: **if** stagnation criterion is satisfied **then**
- 20: Restart worst individuals using opposition-based learning using Equations (31) and (32)
- 21: **end if**
- 22: Record the best fitness value in $cg(t)$
- 23: $t \leftarrow t + 1$
- 24: **end while**
- 25: **return** p_{best}, f_{best}, cg

3.4. Time Complexity

To rigorously characterize the computational cost of SAWG, this section provides a systematic asymptotic analysis of its time complexity. Let N denote the population size, d the problem dimensionality, T the maximum number of iterations, and ENV the computational cost of a single black-box objective evaluation. The computational burden of SAWG mainly arises from four core modules: population initialization, pseudo-gradient

and momentum update, Lévy-flight-based global compensation, and stagnation-aware restart.

Specifically, the initialization stage mainly consists of boundary projection and initial fitness evaluation, yielding a time complexity of $O(N \cdot d + N \cdot ENV)$. In the main iterative stage, pseudo-gradient construction, dimension-wise interaction, and AdamW-based momentum correction introduce an operator-level cost of $O(N \cdot d^2)$; if the individual update process is accompanied by dimension-related black-box evaluations, the corresponding evaluation cost can be expressed as $O(N \cdot d \cdot ENV)$. In addition, Lévy-flight compensation and stagnation-aware restart are conditionally triggered operations, whose worst-case additional costs are $O(N \cdot d + N \cdot ENV)$ and $O(\rho \cdot N \cdot ENV + N \cdot \log N)$, respectively, where ρ denotes the restart ratio.

By aggregating the above modules, the upper bound of the per-iteration time complexity of SAWG can be written as $O(N \cdot d^2 + N \cdot d \cdot ENV)$. Accordingly, the overall time complexity can be further expressed as $O(T \cdot N \cdot d^2 + T \cdot N \cdot d \cdot ENV)$. In typical black-box optimization scenarios, objective evaluation usually dominates the computational cost, i.e., $ENV \gg d$ [1,5], and the above expression can therefore be approximated as $O(T \cdot N \cdot d \cdot ENV)$. This indicates that, although SAWG introduces multiple adaptive evolutionary mechanisms to enhance search quality, its theoretical complexity order still remains within the standard metaheuristic regime dominated by objective evaluations, thereby maintaining a favorable balance between computational cost and search dynamics.

4. Numerical results and analysis on benchmark test suites

4.1. Experimental Study

All numerical experiments were performed under a unified computational environment to ensure reproducibility and fair comparisons. The experimental platform consisted of a 64-bit Windows 11 operating system, an AMD Ryzen 9 7940H processor running at 4.00 GHz with 8 cores and 16 logical threads, and 16 GB of RAM. All algorithms were implemented and visualized using MATLAB R2025b.

To systematically assess the performance of SAWG on classical benchmark optimization problems and to ensure a fair comparison with competing algorithms, experiments were conducted on the CEC2017[30] and CEC2020[31] benchmark suites. In accordance with the official CEC evaluation protocol, each test function was independently run 30 times. All algorithms were executed under identical experimental settings, where the maximum number of function evaluations was fixed at 10,000 and the initial population size was set to 30, thereby ensuring the consistency of the evaluation conditions.

The performance of SAWG was evaluated mainly in terms of the mean and standard deviation over 30 independent runs. The mean was used to characterize the overall optimization performance of the algorithm, whereas the standard deviation was adopted to quantify its stability and robustness. For statistical analysis, Wilcoxon signed-rank tests and Friedman ranking analysis were employed to systematically compare the performance differences among the considered algorithms. In terms of result presentation, SAWG was explicitly highlighted in the convergence curves and boxplots, while the best result for each test function was emphasized in the numerical tables, so as to provide a clear and intuitive comparison of the performance discrepancies among different algorithms.

4.1.1. Parameter Settings and Experimental Protocol

All numerical experiments were performed under a unified computational environment to ensure reproducibility and fair comparisons. The experimental platform consisted of a 64-bit Windows 11 operating system, an AMD Ryzen 9 7940H processor running at

4.00 GHz with 8 cores and 16 logical threads, and 16 GB of RAM. All algorithms were implemented and visualized using MATLAB R2025b.

4.1.2. CEC single-objective benchmark suites

The benchmark suites adopted in this study were released by the IEEE Congress on Evolutionary Computation (CEC). Owing to their complex function landscapes, diverse structural characteristics, and strong capability to characterize the difficulty of high-dimensional optimization problems, these suites have been widely used for performance evaluation of optimization algorithms. In this work, four problem dimensions, namely 10, 30, 50, and 100, were consistently considered on both benchmark suites to systematically assess the performance of SAWG.

After excluding the unimodal function F2, the CEC2017 suite contains 29 test functions, including 2 unimodal functions (F1 and F3), 7 multimodal functions (F4–F10), 10 hybrid functions (F11–F20), and 10 composition functions (F21–F30). The search range of all decision variables is uniformly defined as $[-100, 100]$.

CEC2020 comprises 10 test functions, including one unimodal function (F1), three multimodal functions (F2–F4), three hybrid functions (F5–F7), and three composition functions (F8–F10). These functions were drawn from representative test instances in CEC2014 and CEC2017, and are therefore commonly regarded as a benchmark suite that is both representative and broadly applicable. Given its relatively compact scale and sufficiently comprehensive coverage of function categories, CEC2020 was further employed in this study to analyze the qualitative search behavior, convergence characteristics, and stability of SAWG, thereby avoiding excessive redundancy in figure presentation. By contrast, the sections on optimization accuracy evaluation and statistical testing fully report the more comprehensive experimental results obtained on CEC2017.

4.1.3. Comparison algorithms

To comprehensively evaluate the optimization performance of SAWG, it was compared with nine representative optimization algorithms. The parameter settings of all competing methods followed the recommended configurations reported in their original studies to ensure fairness and comparability.

The first category includes well-established swarm intelligence algorithms, namely particle swarm optimization (PSO)[32], gray wolf optimizer (GWO)[33], whale optimization algorithm (WOA)[34], and Harris hawks optimization (HHO)[35]. These methods generally rely on clear bio-inspired search mechanisms and can achieve a reasonable balance between global exploration and local exploitation, making them widely used in continuous optimization tasks.

The second category consists of recently proposed physics- or nature-inspired metaheuristics, including tornado optimization algorithm (TOC)[36], snow ablation optimizer (SAO)[37], rime optimization algorithm (RIME)[38], phototropism growth algorithm (PGA)[39], and stellar oscillation optimizer (SOO)[40]. These methods are typically constructed from natural-process-inspired modeling and often exhibit strong adaptive search capability, yielding competitive performance on complex optimization problems.

4.2. Qualitative analysis of SAWG

To provide a deeper understanding of the dynamic search behavior of SAWG, a qualitative analysis was conducted on the 10-dimensional CEC2020 benchmark set, as illustrated in 1 – 4 Representative functions from each category were included. Through multiple visualization perspectives, including three-dimensional fitness landscapes, search histories, one-dimensional trajectories, and exploration–exploitation ratios, the search characteristics of the algorithm in various complex spaces were systematically investigated.

It should be noted that the purpose of this qualitative study is to reveal the evolutionary dynamics rather than to pursue ultimate numerical precision. Considering that SAWG exhibits rapid convergence in the early stage, the maximum number of function evaluations in this section was reduced to 2000 in order to avoid severe curve overlap in the later stage and to more clearly present its core evolutionary mechanism.

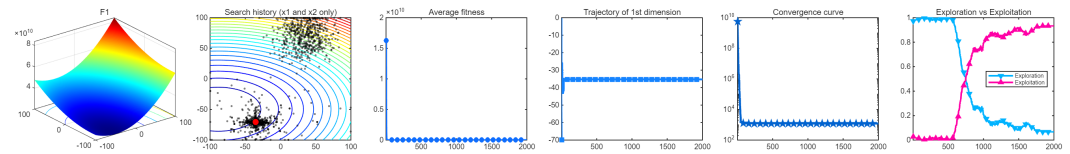


Figure 1. Qualitative analysis of SAWG on Unimodal functions.

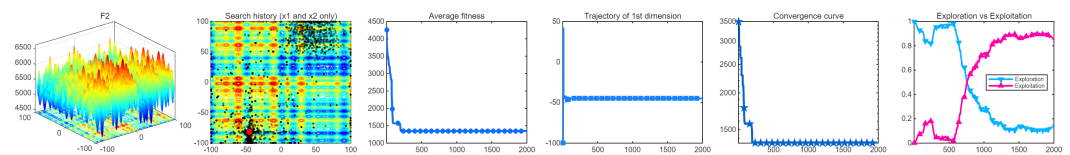


Figure 2. Qualitative analysis of SAWG on Multimodal functions.

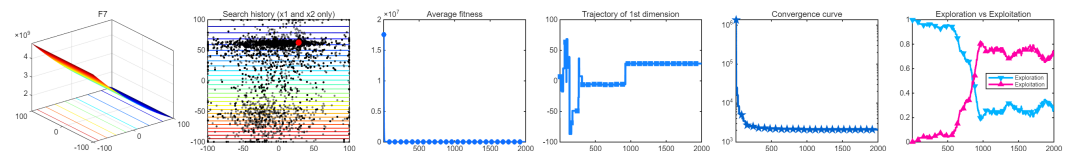


Figure 3. Qualitative analysis of SAWG on Hybrid functions.

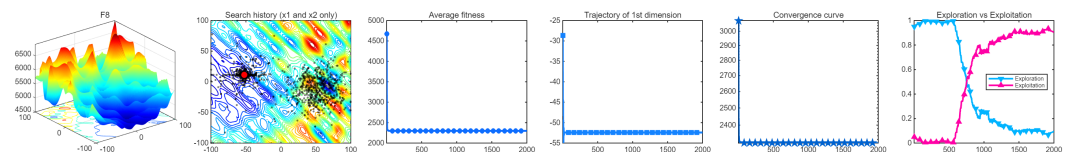


Figure 4. Qualitative analysis of SAWG on Composite functions.

The three-dimensional landscapes and search histories indicate that, under the joint effect of pseudo-gradients and adaptive perturbations, SAWG demonstrates strong structured global coverage in the early stage of optimization, thereby reducing the risk of premature stagnation. The average fitness curves and one-dimensional trajectories further show that SAWG maintains a smooth and stable convergence pattern, where performance improvement is driven more by population-level adaptive learning than by the dominance of a few elite individuals. This characteristic enables the algorithm to transition naturally into the later fine exploitation stage and reduces the likelihood of being trapped in local optima. Meanwhile, the exploration–exploitation ratio curves suggest that SAWG can achieve adaptive transitions between search phases under internal feedback, without relying on predesigned fixed scheduling strategies. Overall, this dynamic collaborative mechanism, which spans extensive early exploration, directional convergence in the middle stage, and refined exploitation in the later stage, provides a mechanistic explanation for the favorable stability and robustness of SAWG on complex landscapes, and also supports its strong quantitative performance on CEC2020.

4.3. Experimental results and discussion on test suite

4.3.1. Numerical Results and Accuracy Analysis

The mean values (Mean) and standard deviations (Std) of 10 algorithms under different dimensional settings on the CEC2017 benchmark suite were analyzed. Among the 116 total

test cases, SAWG ranked first in 97 cases, including 22 best results under the 10D setting and 25 first-place results under each of the 30D, 50D, and 100D settings. These statistics indicate that, as the dimensionality increases, the relative advantage of SAWG is not weakened but instead becomes more pronounced, demonstrating its favorable adaptability to high-dimensional optimization problems. Owing to space limitations, only the results under the 100D setting are presented as a representative example in Table A1.

4.3.2. Convergence Behavior Analysis of SAWG

This study evaluates the convergence[41] performance of SAWG on unimodal, multimodal, hybrid, and composition functions from the CEC2020 benchmark suite. Taking 100 dimensions as an example. As illustrated in Figure 5, SAWG generally exhibits a faster descending trend, smoother convergence trajectories, and higher solution accuracy across different dimensional settings and landscape complexities, and this advantage becomes more pronounced in high-dimensional scenarios.

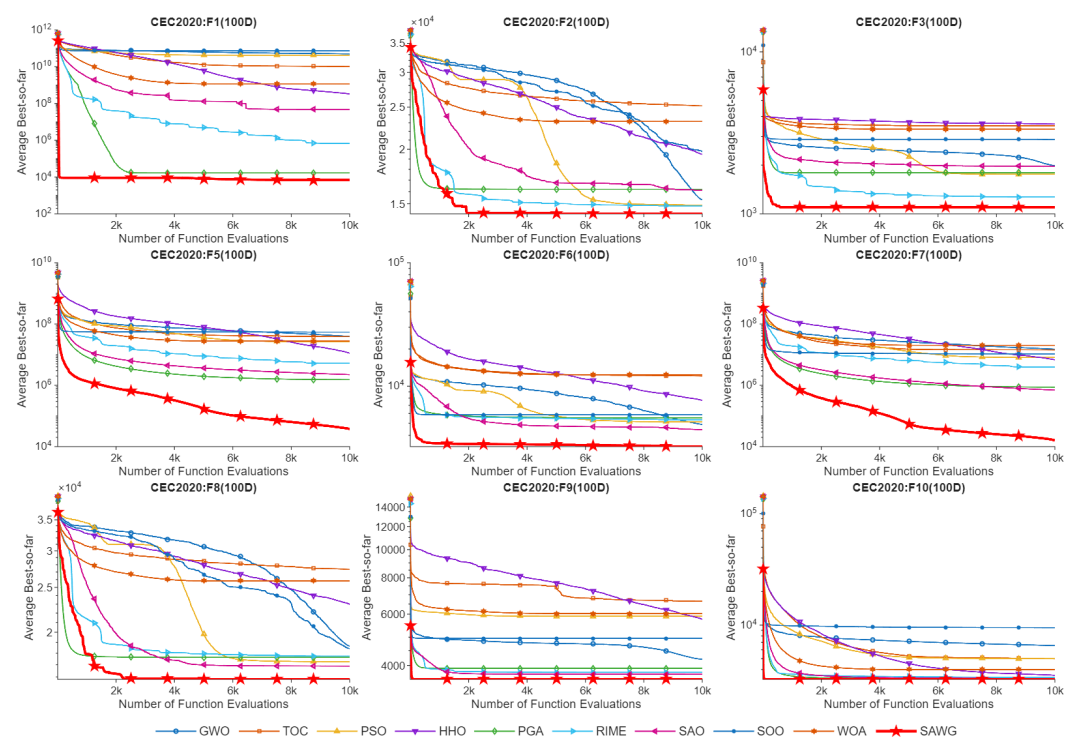


Figure 5. Convergence curves of SAWG and the comparative algorithms on CEC2020 (100-D).

On unimodal functions, SAWG is able to approach the global optimum rapidly with fewer function evaluations, reflecting high directional search efficiency. For multimodal functions, its convergence curves maintain an overall continuous downward trend without obvious premature stagnation, indicating a strong ability to escape from local optima. On the more complex hybrid and composition functions, SAWG can effectively overcome the plateau phases in which most competing algorithms tend to become trapped, while continuing to improve solution quality. From the perspective of convergence behavior, this result supports the positive role of its stagnation detection and adaptive restart mechanisms in nonlinear search spaces. Overall, SAWG demonstrates robust convergence behavior, high solution accuracy, and favorable dimensional scalability.

4.3.3. Box plot analysis of results

During the stability analysis stage, boxplots were utilized to assess the stability of the SAWG algorithm, as shown in Figure 6. Each boxplot corresponds to the same experimental conditions and parameter settings as those used in the convergence plots. In the boxplots,

the blue rectangle represents the box, the whiskers are displayed using blue dashed lines, and outliers are marked with red “+” symbols. The three horizontal lines inside the box represent, from bottom to top, the first quartile Q_1 , the median Q_2 (displayed in red), and the third quartile Q_3 . The interquartile range (IQR) is defined as the height of the box, namely $|Q_3 - Q_1|$. Outliers are identified as data points below $Q_1 - 1.5 IQR$ or above $Q_3 + 1.5 IQR$ [42].

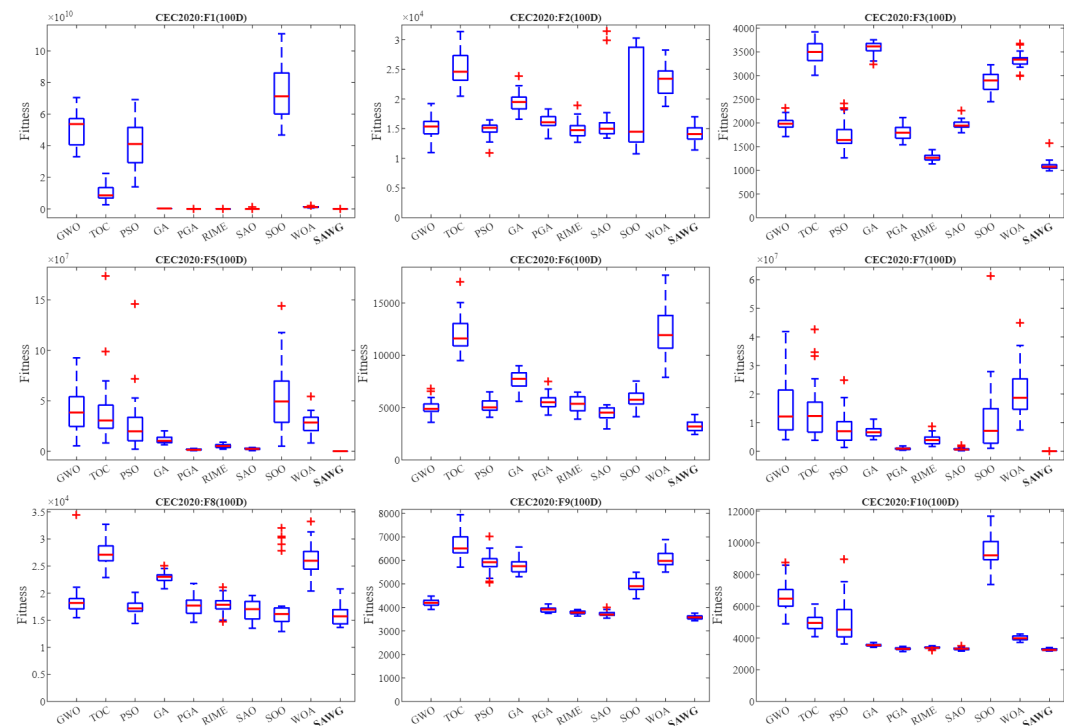


Figure 6. Stability performance of SAWG and the comparative algorithms on CEC2020 (100-D).

The results show that SAWG generally exhibits lower median values and narrower interquartile ranges (IQR) across different test functions and dimensional settings, directly reflecting its superior solution quality and higher concentration of results. Especially in high-dimensional scenarios, the box associated with SAWG becomes noticeably more compact and contains fewer outliers, indicating that it maintains favorable reliability and search consistency over repeated independent runs. Overall, these statistical distribution characteristics suggest that SAWG is able to achieve a favorable balance between optimization accuracy and search stability in complex search spaces. Its advantage is therefore manifested not only in average optimization metrics, but also in the stronger resistance to disturbance reflected by the distribution of its run-level results.

4.4. Statistical tests

On the CEC2017 benchmark suite, the overall and per-function Friedman ranking results of SAWG and the competing algorithms are visualized lollipop radar charts and radar charts, as shown in Section 4.4. In the radar charts, the radial coordinate represents the Friedman ranking of an algorithm on the corresponding test function, where a position closer to the chart center indicates a better ranking. Meanwhile, the size of the enclosed radar region reflects the overall ranking behavior of the algorithm. A smaller region indicates that the algorithm maintains more stable rankings across different functions and achieves a stronger overall performance.

By contrast, the lollipop charts directly present the overall Friedman ranking results on the entire benchmark suite. In these charts, a shorter bar length corresponds to a better

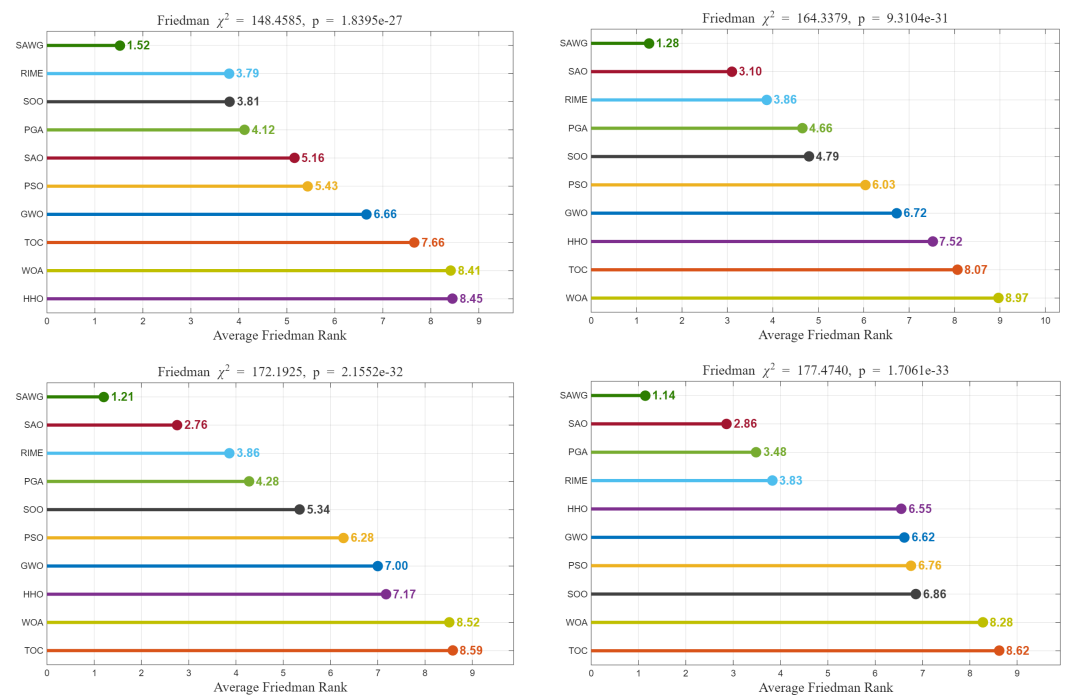


Figure 7. Lollipop charts of SAWG and competing algorithms on the CEC2017 benchmark at 10D, 30D, 50D, and 100D. The top annotation reports the Friedman χ^2 statistic and p-value. A p-values ($p < 0.05$) indicates statistically significant differences among the compared algorithms, while the Friedman χ^2 quantifies the overall extent of ranking differences.

Friedman ranking, thereby providing a concise and intuitive way to compare the global performance of different algorithms.

4.4.1. Friedman Overall Ranking Analysis

The statistical results show that the Friedman tests at all dimensional settings yield extremely small p-values ($p \ll 0.05$), indicating significant performance differences among the compared algorithms. On this basis, SAWG achieves the lowest average Friedman rank across all tested dimensions, showing a clear performance gap over the competing methods. Specifically, in low- and medium-dimensional scenarios, the average rank of SAWG is close to the theoretical optimum of 1. Even in the more challenging high-dimensional cases, its rank changes only slightly and consistently remains in the leading position. As can be seen from the Section 4.4, it is worth noting that from 10dim to 100dim, the Friedman score of SAWG decreased from 1.52 to 1.14, indicating that the algorithm becomes more competitive as the dimensional increases.

4.4.2. Performance Analysis of Friedman Individuals

Taking 100 dimensions as examples. Figure 8 illustrate the per-function Friedman rankings of different algorithms on the CEC2017 test suite using radar charts. The results indicate that the radar region associated with SAWG is overall more compact, with most rankings consistently concentrated near the inner circle, suggesting that SAWG possesses strong overall robustness across different problem structures.

On low-complexity functions and several multimodal functions (e.g., Figure 8), SAWG generally maintains the best or second-best ranking, outperforming competing algorithms such as GWO and PSO that exhibit larger fluctuations. This indicates that SAWG can establish effective search directions rapidly while maintaining a favorable balance between convergence efficiency and search stability. On more complex multimodal functions, the rankings of most competing algorithms expand noticeably outward, whereas SAWG

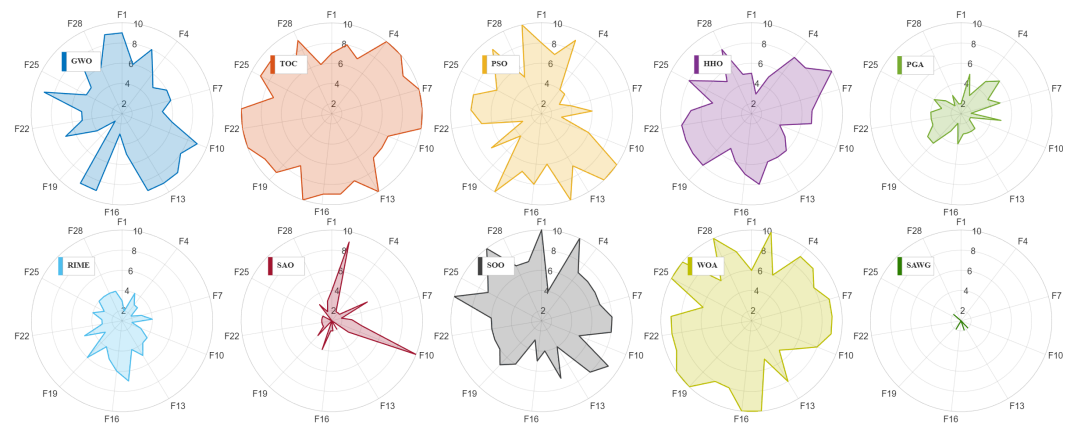


Figure 8. Radar charts of the CEC2017 test suite under 100D. As the benchmark includes 29 test functions, only one label is shown for every three functions on the radar charts for better readability.

still preserves a relatively stable inner-circle distribution, suggesting that it can maintain desirable search stability even in complicated search spaces.

Overall, these statistical results suggest that the performance advantage of SAWG does not rely on specific function types or dimensional settings, but is instead reflected in stable and broadly applicable optimization performance across diverse test scenarios.

4.4.3. Wilcoxon signed-rank test

To systematically assess the performance advantage of SAWG, Tables 1 and 2 summarize the results of the Wilcoxon signed-rank test and Holm multiple-comparison correction on the CEC2017 and CEC2022 benchmark suites, respectively. The significance level was uniformly set to 0.05, where “+”, “≈”, and “-” indicate significant superiority, no significant difference, and significant inferiority, respectively.

The statistical results show that, on unimodal functions where algorithmic performance tends to be relatively similar, SAWG generally maintains performance comparable to or slightly better than that of mainstream algorithms. By contrast, in more challenging optimization scenarios characterized by high dimensionality, nonconvexity, and multimodality, SAWG obtains a larger number of statistically significant advantages (“+”). This suggests that the core search mechanism of SAWG possesses favorable scalability and stability when handling complex nonlinear search spaces. However, it is undeniable that recent new algorithms (such as SOO, PGA, RIME, etc.) are also quite competitive in low-dimensional environments. Overall, these statistical tests not only support the competitive advantage of SAWG in terms of overall optimization performance, but also indicate, from a statistical perspective, that its performance gains remain relatively stable and consistent across different problem structures and dimensional settings.

5. Application of SAWG to Real-world Problems

This section compares SAWG with nine competing algorithms on eight real-world constrained engineering optimization problems, aiming to demonstrate the effectiveness and robustness of SAWG in practical engineering scenarios.

To assess the applicability of SAWG to practical constrained optimization, this study further considers eight classical engineering design problems, including the tension/compression spring[40,43], pressure vessel[36,40], welded beam[36,38], gear train[44], rolling element bearing[45], cantilever beam[36,39], planetary gear train[46], and robot gripper problems[46]. These tasks cover continuous, discrete, and mixed decision variables, while simultaneously involving nonlinear, inequality, equality, and strongly coupled constraints. They therefore provide a systematic testbed for evaluating algorithmic perfor-

Table 1. Wilcoxon rank-sum test (with Holm correction) results summary on the CEC2017 test functions.

CEC2017	Type	Pairwise comparison results (+ / $\approx$ / −)								
		GWO	TOC	PSO	HHO	PGA	RIME	SAO	SOO	WOA
10Dim	Unimodal	2/0/0	2/0/0	1/0/1	2/0/0	1/0/1	2/0/0	1/0/1	0/0/2	2/0/0
	Multimodal	7/0/0	7/0/0	3/2/2	7/0/0	6/1/0	6/1/0	5/1/1	4/3/0	7/0/0
	Hybrid	10/0/0	9/1/0	9/1/0	10/0/0	9/1/0	8/2/0	9/1/0	9/0/1	10/0/0
	Composite	8/2/0	7/3/0	8/2/0	8/2/0	6/3/1	7/3/0	6/4/0	6/4/0	8/2/0
	Total	27/2/0	25/4/0	21/5/3	27/2/0	22/5/2	23/6/0	21/6/2	19/7/3	27/2/0
30Dim	Unimodal	2/0/0	2/0/0	2/0/0	2/0/0	2/0/0	2/0/0	2/0/0	2/0/0	2/0/0
	Multimodal	6/1/0	7/0/0	6/1/0	7/0/0	7/0/0	6/0/1	6/1/0	6/1/0	7/0/0
	Hybrid	10/0/0	10/0/0	10/0/0	10/0/0	9/1/0	8/2/0	7/3/0	8/2/0	10/0/0
	Composite	10/0/0	10/0/0	10/0/0	10/0/0	8/2/0	10/0/0	7/3/0	10/0/0	10/0/0
	Total	28/1/0	29/0/0	28/1/0	29/0/0	26/3/0	26/2/1	22/7/0	26/3/0	29/0/0
50Dim	Unimodal	2/0/0	2/0/0	2/0/0	2/0/0	2/0/0	2/0/0	1/1/0	2/0/0	2/0/0
	Multimodal	6/1/0	7/0/0	6/1/0	7/0/0	7/0/0	6/1/0	6/1/0	6/1/0	7/0/0
	Hybrid	10/0/0	10/0/0	10/0/0	10/0/0	8/2/0	9/1/0	8/2/0	9/1/0	10/0/0
	Composite	10/0/0	10/0/0	9/1/0	10/0/0	8/2/0	8/2/0	7/3/0	9/1/0	10/0/0
	Total	28/1/0	29/0/0	27/2/0	29/0/0	25/4/0	25/4/0	22/7/0	26/3/0	29/0/0
100Dim	Unimodal	2/0/0	2/0/0	2/0/0	2/0/0	2/0/0	2/0/0	2/0/0	2/0/0	2/0/0
	Multimodal	7/0/0	7/0/0	6/1/0	7/0/0	7/0/0	6/1/0	7/0/0	6/1/0	7/0/0
	Hybrid	10/0/0	10/0/0	10/0/0	10/0/0	7/3/0	10/0/0	7/3/0	10/0/0	10/0/0
	Composite	10/0/0	10/0/0	10/0/0	10/0/0	10/0/0	10/0/0	9/1/0	9/1/0	10/0/0
	Total	29/0/0	29/0/0	28/1/0	29/0/0	26/3/0	28/1/0	25/4/0	27/2/0	29/0/0

mance from multiple perspectives, including feasible-region preservation, global search capability, and solution stability. Due to space limitations, only the mathematical formulation of the pressure vessel design problem is presented in the main text. As shown in Equations (39)–(41). This problem aims to jointly optimize the inner radius R , head thickness T_h , cylindrical section length L , and shell thickness T_s under structural safety and manufacturing requirements so as to minimize the total manufacturing cost. Although the remaining engineering problems are not described mathematically in detail, their optimization structures collectively cover narrow feasible regions, variable sensitivity, discrete combinatorial characteristics, and high-dimensional constraint coupling that are commonly encountered in constrained engineering design. They can therefore still serve as effective benchmarks for evaluating the engineering generalization capability of SAWG.

Tables 3–10 show that, in the tension/compression spring design problem, SAWG achieves an average objective value of 1.273E-02, matching the best-performing competing methods such as SOO and PGA. Meanwhile, the convergence curves and boxplots indicate that SAWG can enter high-quality feasible regions at an early stage and maintain stable convergence with relatively low fluctuation. Table 4 shows that, in the pressure vessel design problem, SAWG attains the best observed value of 7.535E+02 with an extremely small standard deviation, indicating that it can stably produce high-quality feasible solutions under strict constraints. Table 5 further shows that, in the welded beam design problem, SAWG simultaneously achieves the best mean value of 3.482E-10 and the best observed value of 2.701E-12, reflecting its high solution accuracy under narrow feasible regions and strongly coupled constraints. Table 6 indicates that, for the gear train design problem, which is a discrete multimodal task, SAWG consistently maintains competitive average performance and stably obtains the best solution, suggesting that its search mecha-

Table 2. Wilcoxon rank-sum test (with Holm correction) results summary on the CEC2020 test functions.

CEC2020	Type	Pairwise comparison results (+ / $\approx$ / -)								
		GWO	TOC	PSO	HHO	PGA	RIME	SAO	SOO	WOA
10Dim	Unimodal	1/0/0	1/0/0	1/0/0	1/0/0	1/0/0	1/0/0	1/0/0	0/0/1	1/0/0
	Multimodal	2/0/1	3/0/0	3/0/0	2/0/1	3/0/0	1/2/0	3/0/0	0/2/1	2/0/1
	Hybrid	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0
	Composite	2/1/0	2/1/0	2/1/0	2/1/0	2/1/0	2/1/0	2/1/0	2/1/0	2/1/0
	Total	8/1/1	9/1/0	9/1/0	8/1/1	9/1/0	7/3/0	9/1/0	5/3/2	8/1/1
30Dim	Unimodal	1/0/0	1/0/0	1/0/0	1/0/0	1/0/0	1/0/0	1/0/0	1/0/0	1/0/0
	Multimodal	2/0/1	3/0/0	3/0/0	2/0/1	3/0/0	2/1/0	3/0/0	1/0/2	2/0/1
	Hybrid	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0
	Composite	3/0/0	3/0/0	3/0/0	3/0/0	2/1/0	3/0/0	1/2/0	3/0/0	3/0/0
	Total	9/0/1	10/0/0	10/0/0	9/0/1	9/1/0	9/1/0	8/2/0	8/0/2	9/0/1
50Dim	Unimodal	1/0/0	1/0/0	1/0/0	1/0/0	1/0/0	1/0/0	0/1/0	1/0/0	1/0/0
	Multimodal	2/0/1	2/0/1	3/0/0	2/0/1	3/0/0	2/1/0	2/1/0	1/1/1	2/0/1
	Hybrid	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0
	Composite	3/0/0	3/0/0	2/1/0	3/0/0	2/1/0	1/2/0	1/2/0	2/1/0	3/0/0
	Total	9/0/1	9/0/1	9/1/0	9/0/1	9/1/0	7/3/0	6/4/0	7/2/1	9/0/1
100Dim	Unimodal	1/0/0	1/0/0	1/0/0	1/0/0	1/0/0	1/0/0	1/0/0	1/0/0	1/0/0
	Multimodal	2/0/1	2/1/0	3/0/0	2/0/1	3/0/0	2/1/0	3/0/0	2/0/1	2/0/1
	Hybrid	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0
	Composite	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	3/0/0	2/1/0	3/0/0
	Total	9/0/1	9/1/0	10/0/0	9/0/1	10/0/0	9/1/0	10/0/0	8/1/1	9/0/1

nism remains highly repeatable under combinatorial structures. Table 7 shows that, in the rolling element bearing design problem, the mean and best values obtained by SAWG are identical and the variance is zero, indicating strong stability on high-dimensional problems with strict constraints. Table 8 shows that, in the cantilever beam design problem, SAWG achieves one of the best objective values with extremely small variance, indicating that it can still maintain effective control near flat optimal regions. Table 9 indicates that, in the planetary gear train design problem, SAWG achieves the best results in both mean objective value and worst-case performance, while converging stably to a high-quality solution. Table 10 further shows that, in the robot gripper problem, SAWG obtains the lowest mean objective value and a near-zero best value with extremely small variance. Taken together, these results suggest that the advantage of SAWG does not rely on any single problem structure, but is instead reflected in the coordinated enhancement of feasibility maintenance, search stability, and solution quality across different variable types, constraint forms, and objective landscapes.

Objective function:

$$\min_{\mathbf{x}} f(\mathbf{x}) = 0.6224 x_1 x_3 x_4 + 1.7781 x_2 x_3^2 + 3.1661 x_1^2 x_4 + 19.84 x_1^2 x_3, \quad (39)$$

where

$$\mathbf{x} = [x_1, x_2, x_3, x_4] = [T_s, T_h, R, L].$$

Constraint:

$$\begin{aligned}
 g_1(\mathbf{x}) &= -x_1 + 0.0193x_3 \leq 0, \\
 g_2(\mathbf{x}) &= -x_2 + 0.00954x_3 \leq 0, \\
 g_3(\mathbf{x}) &= -\pi x_3^2 x_4 - \frac{4}{3}\pi x_3^3 + 1296000 \leq 0, \\
 g_4(\mathbf{x}) &= x_4 - 240 \leq 0.
 \end{aligned} \tag{40}$$

Variable range:

$$\begin{aligned}
 0.05 &\leq x_1 \leq 2.00, \\
 0.25 &\leq x_2 \leq 1.30, \\
 2.00 &\leq x_3 \leq 15.00.
 \end{aligned} \tag{41}$$

Table 3. Tension spring design problem.

	GWO	TOC	PSO	HHO	PGA	RIME	SAO	SOO	WOA	SAWG
Mean	1.281E-02	1.394E-02	1.321E-02	1.352E-02	1.292E-02	1.618E-02	1.342E-02	1.273E-02	1.351E-02	1.273E-02
Std	1.304E-04	4.493E-03	9.340E-04	8.643E-04	5.866E-04	2.033E-03	1.201E-03	7.613E-05	1.029E-03	7.793E-05
Best	1.269E-02	1.267E-02	1.267E-02	1.270E-02	1.267E-02	1.274E-02	1.268E-02	1.267E-02	1.267E-02	1.267E-02
Worst	1.322E-02	3.045E-02	1.777E-02	1.642E-02	1.587E-02	1.802E-02	1.774E-02	1.296E-02	1.639E-02	1.312E-02
X1	1.281E-02	1.394E-02	1.321E-02	1.352E-02	1.292E-02	1.618E-02	1.342E-02	1.273E-02	1.351E-02	1.273E-02
X2	1.304E-04	4.493E-03	9.340E-04	8.643E-04	5.866E-04	2.033E-03	1.201E-03	7.613E-05	1.029E-03	7.793E-05
X3	1.269E-02	1.267E-02	1.267E-02	1.270E-02	1.267E-02	1.274E-02	1.268E-02	1.267E-02	1.267E-02	1.267E-02

Table 4. Pressure vessel design problem.

	GWO	TOC	PSO	HHO	PGA	RIME	SAO	SOO	WOA	SAWG
Mean	8.035E+02	8.028E+02	7.535E+02	1.272E+03	7.535E+02	7.558E+02	9.259E+02	7.535E+02	8.362E+02	7.535E+02
Std	1.873E+02	1.875E+02	1.156E-13	2.995E+02	1.156E-13	7.750E+00	3.179E+02	1.156E-13	2.253E+02	4.483E-05
Best	7.535E+02	7.535E+02	7.535E+02	7.537E+02	7.535E+02	7.535E+02	7.535E+02	7.535E+02	7.535E+02	7.535E+02
Worst	1.493E+03	1.493E+03	7.535E+02	1.559E+03	7.535E+02	7.959E+02	1.493E+03	7.535E+02	1.503E+03	7.535E+02
X1	8.035E+02	8.028E+02	7.535E+02	1.272E+03	7.535E+02	7.558E+02	9.259E+02	7.535E+02	8.362E+02	7.535E+02
X2	1.873E+02	1.875E+02	1.156E-13	2.995E+02	1.156E-13	7.750E+00	3.179E+02	1.156E-13	2.253E+02	4.483E-05
X3	7.535E+02	7.535E+02	7.535E+02	7.537E+02	7.535E+02	7.535E+02	7.535E+02	7.535E+02	7.535E+02	7.535E+02
X4	1.493E+03	1.493E+03	7.535E+02	1.559E+03	7.535E+02	7.959E+02	1.493E+03	7.535E+02	1.503E+03	7.535E+02

Table 5. Welded beam design problem.

	GWO	TOC	PSO	HHO	PGA	RIME	SAO	SOO	WOA	SAWG
Mean	5.185E-10	3.255E-09	3.894E-09	4.435E-09	1.254E-09	2.054E-09	7.074E-09	7.810E-10	3.568E-09	3.482E-10
Std	5.073E-10	4.243E-09	6.543E-09	7.895E-09	8.742E-10	4.880E-09	8.768E-09	7.994E-10	6.890E-09	4.543E-10
Best	2.701E-12	2.701E-12	2.701E-12	2.308E-11	2.308E-11	2.308E-11	2.308E-11	2.701E-12	2.701E-12	2.701E-12
Worst	1.362E-09	1.827E-08	2.726E-08	2.726E-08	3.300E-09	2.726E-08	2.726E-08	2.358E-09	2.726E-08	1.362E-09
X1	5.185E-10	3.255E-09	3.894E-09	4.435E-09	1.254E-09	2.054E-09	7.074E-09	7.810E-10	3.568E-09	3.482E-10
X2	5.073E-10	4.243E-09	6.543E-09	7.895E-09	8.742E-10	4.880E-09	8.768E-09	7.994E-10	6.890E-09	4.543E-10
X3	2.701E-12	2.701E-12	2.701E-12	2.308E-11	2.308E-11	2.308E-11	2.308E-11	2.701E-12	2.701E-12	2.701E-12
X4	1.362E-09	1.827E-08	2.726E-08	2.726E-08	3.300E-09	2.726E-08	2.726E-08	2.358E-09	2.726E-08	1.362E-09

Finally, based on the above eight typical constrained engineering optimization problems, Friedman ranking analyses were conducted on the mean and standard deviation results of SAWG and the competing algorithms, as illustrated in Figures 9 and 10. SAWG consistently maintained the highest Friedman score and had the smallest radar chart area. This shows that SAWG simultaneously achieves superior solution quality, higher stability, and stronger cross-problem generalization ability across a variety of heterogeneous constrained engineering optimization tasks. This indicates that SAWG possesses favorable effectiveness and application potential in practical engineering optimization scenarios.

Table 6. Gear train design problem.

	GWO	TOC	PSO	HHO	PGA	RIME	SAO	SOO	WOA	SAWG
Mean	8.035E+02	8.028E+02	7.535E+02	1.272E+03	7.535E+02	7.558E+02	9.259E+02	7.535E+02	8.362E+02	7.535E+02
Std	1.873E+02	1.875E+02	1.156E-13	2.995E+02	1.156E-13	7.750E+00	3.179E+02	1.156E-13	2.253E+02	4.483E-05
Best	7.535E+02	7.535E+02	7.535E+02	7.537E+02	7.535E+02	7.535E+02	7.535E+02	7.535E+02	7.535E+02	7.535E+02
Worst	1.493E+03	1.493E+03	7.535E+02	1.559E+03	7.535E+02	7.959E+02	1.493E+03	7.535E+02	1.503E+03	7.535E+02
X1	8.035E+02	8.028E+02	7.535E+02	1.272E+03	7.535E+02	7.558E+02	9.259E+02	7.535E+02	8.362E+02	7.535E+02
X2	1.873E+02	1.875E+02	1.156E-13	2.995E+02	1.156E-13	7.750E+00	3.179E+02	1.156E-13	2.253E+02	4.483E-05
X3	7.535E+02	7.535E+02	7.535E+02	7.537E+02	7.535E+02	7.535E+02	7.535E+02	7.535E+02	7.535E+02	7.535E+02
X4	1.493E+03	1.493E+03	7.535E+02	1.559E+03	7.535E+02	7.959E+02	1.493E+03	7.535E+02	1.503E+03	7.535E+02

Table 7. Rolling element bearing design.

	GWO	TOC	PSO	HHO	PGA	RIME	SAO	SOO	WOA	SAWG
Mean	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.040E+05	-3.047E+05
Std	6.000E+00	3.600E+01	2.800E+01	1.700E+01	0.000E+00	2.000E+00	0.000E+00	0.000E+00	3.967E+03	0.000E+00
Best	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05
Worst	-3.047E+05	-3.046E+05	-3.046E+05	-3.046E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-2.830E+05	-3.047E+05
X1	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.040E+05	-3.047E+05
X2	6.000E+00	3.600E+01	2.800E+01	1.700E+01	0.000E+00	2.000E+00	0.000E+00	0.000E+00	3.967E+03	0.000E+00
X3	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05
X4	-3.047E+05	-3.046E+05	-3.046E+05	-3.046E+05	-3.047E+05	-3.047E+05	-3.047E+05	-3.047E+05	-2.830E+05	-3.047E+05
X5	1.250E+02	1.250E+02	1.250E+02	1.250E+02	1.250E+02	1.250E+02	1.250E+02	1.250E+02	1.250E+02	1.250E+02
X6	2.586E+01	2.586E+01	2.586E+01	2.586E+01	2.586E+01	2.585E+01	2.586E+01	2.586E+01	2.584E+01	2.586E+01
X7	5.000E+01	5.000E+01	5.000E+01	5.000E+01	5.000E+01	5.000E+01	5.000E+01	5.000E+01	5.000E+01	5.000E+01
X8	5.150E-01	5.150E-01	5.150E-01	5.150E-01	5.150E-01	5.150E-01	5.150E-01	5.150E-01	5.150E-01	5.150E-01
X9	5.185E-01	5.460E-01	5.292E-01	5.198E-01	5.150E-01	5.150E-01	5.150E-01	5.150E-01	5.428E-01	5.150E-01
X10	4.479E-01	4.600E-01	4.499E-01	4.524E-01	4.468E-01	4.329E-01	4.563E-01	4.459E-01	4.332E-01	4.384E-01

Table 8. Cantilever beam design problem.

	GWO	TOC	PSO	HHO	PGA	RIME	SAO	SOO	WOA	SAWG
Mean	1.340E+00	1.343E+00	1.340E+00	1.344E+00	1.340E+00	1.376E+00	1.340E+00	1.340E+00	1.637E+00	1.340E+00
Std	5.962E-05	5.382E-03	2.629E-06	2.096E-03	1.061E-04	5.758E-02	1.866E-09	1.989E-12	1.753E-01	1.733E-07
Best	1.340E+00	1.340E+00	1.340E+00	1.340E+00	1.340E+00	1.342E+00	1.340E+00	1.340E+00	1.405E+00	1.340E+00
Worst	1.340E+00	1.357E+00	1.340E+00	1.348E+00	1.340E+00	1.658E+00	1.340E+00	1.340E+00	2.133E+00	1.340E+00
X1	1.340E+00	1.343E+00	1.340E+00	1.344E+00	1.340E+00	1.376E+00	1.340E+00	1.340E+00	1.637E+00	1.340E+00
X2	5.962E-05	5.382E-03	2.629E-06	2.096E-03	1.061E-04	5.758E-02	1.866E-09	1.989E-12	1.753E-01	1.733E-07
X3	1.340E+00	1.340E+00	1.340E+00	1.340E+00	1.340E+00	1.342E+00	1.340E+00	1.340E+00	1.405E+00	1.340E+00
X4	1.340E+00	1.357E+00	1.340E+00	1.348E+00	1.340E+00	1.658E+00	1.340E+00	1.340E+00	2.133E+00	1.340E+00
X5	6.015E+00	6.000E+00	6.015E+00	6.008E+00	6.009E+00	6.204E+00	6.016E+00	6.016E+00	6.285E+00	6.016E+00

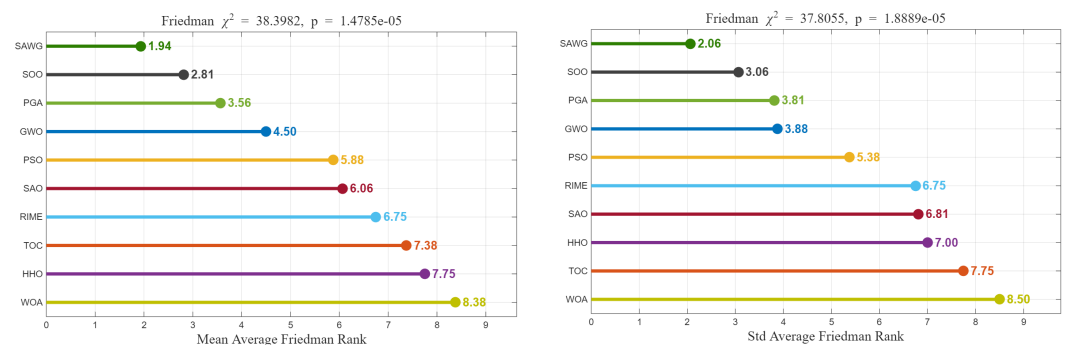
Table 9. Planetary Gear Train Design.

	GWO	TOC	PSO	HHO	PGA	RIME	SAO	SOO	WOA	SAWG
Mean	2.412E-01	6.346E+01	7.168E+01	2.397E-01	2.641E-01	2.416E-01	1.800E+03	2.416E-01	2.482E-01	2.394E-01
Std	4.067E-03	3.454E+02	3.911E+02	2.156E-03	5.932E-02	6.140E-03	3.662E+03	8.372E-03	1.750E-02	3.758E-03
Best	2.353E-01	2.355E-01	2.368E-01	2.359E-01	2.378E-01	2.355E-01	2.394E-01	2.353E-01	2.359E-01	2.350E-01
Worst	2.500E-01	1.892E+03	2.142E+03	2.445E-01	5.645E-01	2.689E-01	9.000E+03	2.782E-01	3.346E-01	2.485E-01
X1	2.412E-01	6.346E+01	7.168E+01	2.397E-01	2.641E-01	2.416E-01	1.800E+03	2.416E-01	2.482E-01	2.394E-01
X2	4.067E-03	3.454E+02	3.911E+02	2.156E-03	5.932E-02	6.140E-03	3.662E+03	8.372E-03	1.750E-02	3.758E-03
X3	2.353E-01	2.355E-01	2.368E-01	2.359E-01	2.378E-01	2.355E-01	2.394E-01	2.353E-01	2.359E-01	2.350E-01
X4	2.500E-01	1.892E+03	2.142E+03	2.445E-01	5.645E-01	2.689E-01	9.000E+03	2.782E-01	3.346E-01	2.485E-01
X5	3.588E+01	2.827E+01	4.436E+01	2.529E+01	3.360E+01	3.795E+01	1.840E+01	3.826E+01	2.354E+01	3.027E+01
X6	2.520E+01	1.989E+01	2.470E+01	2.295E+01	2.177E+01	2.528E+01	1.629E+01	2.640E+01	2.112E+01	2.260E+01
X7	1.928E+01	1.897E+01	1.617E+01	1.990E+01	1.740E+01	1.837E+01	1.430E+01	1.971E+01	1.777E+01	1.658E+01
X8	3.033E+01	2.656E+01	3.032E+01	2.517E+01	2.804E+01	2.995E+01	1.778E+01	3.133E+01	2.224E+01	2.520E+01
X9	1.781E+01	1.585E+01	1.669E+01	1.998E+01	1.623E+01	1.718E+01	1.424E+01	1.871E+01	1.733E+01	1.539E+01

Taken together, the numerical benchmark experiments and engineering case studies demonstrate that SAWG is able to maintain a favorable balance between exploration and exploitation across different optimization scenarios, thereby exhibiting strong competitiveness and general applicability. Meanwhile, SAWG consistently preserves stable convergence

Table 10. Robot Gripper Problem.

	GWO	TOC	PSO	HHO	PGA	RIME	SAO	SOO	WOA	SAWG
Mean	2.977E+00	3.106E+00	3.370E+00	1.309E+01	1.012E-16	3.020E+00	1.690E+00	1.969E+00	3.479E+00	7.568E-17
Std	1.249E+00	2.434E+00	6.033E-01	2.188E+01	3.616E-17	2.246E+00	1.946E+00	1.430E+00	3.162E+00	8.888E-18
Best	7.346E-17	8.074E-17	2.547E+00	1.486E-16	7.274E-17	7.344E-17	7.274E-17	1.064E-16	1.456E-16	7.274E-17
Worst	4.095E+00	1.129E+01	4.314E+00	8.051E+01	1.797E-16	1.036E+01	5.179E+00	3.658E+00	9.292E+00	1.128E-16
X1	2.977E+00	3.106E+00	3.370E+00	1.309E+01	1.012E-16	3.020E+00	1.690E+00	1.969E+00	3.479E+00	7.568E-17
X2	1.249E+00	2.434E+00	6.033E-01	2.188E+01	3.616E-17	2.246E+00	1.946E+00	1.430E+00	3.162E+00	8.888E-18
X3	7.346E-17	8.074E-17	2.547E+00	1.486E-16	7.274E-17	7.344E-17	7.274E-17	1.064E-16	1.456E-16	7.274E-17
X4	4.095E+00	1.129E+01	4.314E+00	8.051E+01	1.797E-16	1.036E+01	5.179E+00	3.658E+00	9.292E+00	1.128E-16
X5	1.443E+02	1.390E+02	1.491E+02	1.379E+02	9.702E+01	1.326E+02	1.225E+02	1.294E+02	1.232E+02	9.987E+01
X6	1.229E+02	1.149E+02	1.193E+02	1.069E+02	3.521E+01	9.375E+01	6.989E+01	9.435E+01	7.511E+01	3.806E+01
X7	1.868E+02	1.801E+02	1.893E+02	1.428E+02	1.920E+02	1.900E+02	1.766E+02	1.900E+02	1.392E+02	1.960E+02

**Figure 9.** Engineering Test Cases Lollipop Chart.**Figure 10.** Radar charts of engineering test cases. TC: tension/compression spring; PV: pressure vessel; WB: welded beam; GT: gear train; REB: rolling element bearing; CB: cantilever beam; PGT: planetary gear train; RG: robot gripper.

characteristics and high-quality solutions under diverse heterogeneous problem settings, further suggesting that its performance advantage is not limited to standard benchmark tests but extends to a certain degree of cross-scenario generalization. Overall, SAWG not only achieves strong performance on theoretical benchmark problems, but also shows a favorable capability for handling complex real-world engineering design tasks.

6. Discussion and Future Directions

6.1. Conclusions

In this study, a novel adaptive gradient-guided metaheuristic optimization algorithm, termed Self-Adaptive AdamW-guided Optimization (SAWG), is proposed to address the core challenges commonly encountered in complex black-box optimization, including the unavailability of gradient information and the inherent trade-off between search efficiency and stability. Unlike conventional swarm intelligence algorithms that primarily rely on

purely random operators or empirical heuristic rules, SAWG introduces gradient-inspired guidance while preserving the generality and robustness of metaheuristic optimization, thereby systematically integrating gradient-like learning mechanisms into population-based search.

The key characteristics of SAWG can be summarized as follows. First, to cope with the absence of true gradient information in black-box optimization, a population-statistics-based pseudo-gradient estimation mechanism is constructed, providing stable and computable directional guidance for the search process. Second, an AdamW-style adaptive update strategy is incorporated into the population evolution, embedding first- and second-order moment estimation together with weight decay regularization to dynamically regulate both search direction and step size. Third, a stagnation-aware adaptive control mechanism is designed to identify potential efficiency degradation during the search process and to effectively alleviate premature convergence through targeted exploration compensation. These mechanisms operate synergistically within a unified framework, enabling SAWG to achieve an adaptive balance between exploration and exploitation.

To systematically evaluate the performance of SAWG, qualitative analyses are first conducted, demonstrating that SAWG exhibits more reasonable and stable behavioral characteristics in terms of search trajectory structure, smoothness of fitness evolution, and dynamic exploration–exploitation regulation, thereby validating the effectiveness of its gradient-guided update mechanism. Subsequently, extensive experiments are carried out on representative and widely adopted CEC benchmark suites, including CEC2017 and CEC2020, under multiple dimensional settings. The experimental results indicate that SAWG consistently achieves significant or highly competitive performance advantages over a range of strong and competitive comparison algorithms across multiple evaluation criteria, including best and mean fitness values, convergence behavior, stability, and statistical significance.

Furthermore, to assess its practical applicability in real-world scenarios, SAWG is successfully applied to several classical engineering design optimization problems. The results demonstrate that, even in the presence of nonlinear constraints, complex feasible regions, and strong variable coupling, SAWG can reliably obtain high-quality solutions while maintaining consistent advantages in both accuracy and robustness. These findings clearly indicate that SAWG is not only competitive on theoretical benchmark problems but also exhibits strong generalization capability and application potential in practical engineering optimization.

6.2. Future Work

Although SAWG has demonstrated encouraging performance, there remains considerable room for further extension and enhancement. Future research will focus on the following directions:

- Algorithmic enhancement: further improving pseudo-gradient construction and adaptive update strategies, for example by incorporating more locally sensitive population information or multi-scale learning mechanisms, to enhance search efficiency in high-dimensional and highly complex problems.
- Theoretical analysis: conducting more in-depth investigations into the search dynamics and convergence properties of SAWG, in order to systematically reveal the underlying mechanisms by which gradient-inspired guidance promotes a balance between exploration and exploitation in metaheuristic search.
- Extended application scenarios: extending SAWG to multi-objective, large-scale, and dynamic optimization problems, in conjunction with more advanced constraint-

handling techniques, to further broaden its applicability to complex real-world problems.

In summary, SAWG provides a new paradigm for the systematic integration of gradient learning concepts and metaheuristic optimization. By introducing adaptive and learnable search mechanisms into gradient-free environments, SAWG offers a robust and efficient solution for both complex benchmark functions and practical engineering optimization problems, while also serving as a valuable reference for future research on learning-driven metaheuristic algorithms.

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Data Availability Statement: The data used in this study must be authentic and reliable to ensure the validity and reproducibility of the experimental results. The datasets and source code used in this study are openly available at: <https://github.com/xieeix-code/The-Intelligent-Optimization-Algorithm-SAWG> (accessed on January 27, 2026). All benchmark data sets used in this study (CEC2017 and CEC2020) are publicly available from the official CEC competition website.

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Table A1. Running results of 10 algorithms in the CEC2017 (100-D)

Function		GWO	TOC	PSO	HHO	PGA	RIME	SAO	SOO	WOA	SAWG
F_1	Mean	5.02E+10	1.02E+10	4.26E+10	3.27E+08	1.66E+04	6.78E+05	4.79E+07	7.44E+10	1.17E+09	6.97E+03
	Std	1.10E+10	4.76E+09	1.57E+10	3.63E+07	1.83E+04	4.01E+05	2.62E+08	1.63E+10	3.46E+08	1.01E+04
F_3	Mean	2.22E+05	2.65E+05	2.63E+05	1.37E+05	2.08E+05	9.35E+04	7.31E+05	1.89E+05	7.59E+05	3.00E+02
	Std	1.84E+04	1.33E+05	8.65E+04	1.80E+04	2.14E+04	1.77E+04	1.41E+05	3.78E+04	1.38E+05	8.86E-08
F_4	Mean	4.85E+03	2.47E+03	5.79E+03	9.08E+02	6.47E+02	7.27E+02	6.46E+02	9.53E+03	1.52E+03	4.85E+02
	Std	1.60E+03	5.85E+02	3.61E+03	6.81E+01	3.46E+01	5.86E+01	2.95E+01	2.69E+03	2.00E+02	9.97E+01
F_5	Mean	1.15E+03	1.73E+03	1.02E+03	1.45E+03	1.12E+03	9.75E+02	9.24E+02	1.18E+03	1.51E+03	7.16E+02
	Std	6.70E+01	1.10E+02	8.31E+01	5.27E+01	9.83E+01	8.07E+01	1.23E+02	6.27E+01	8.76E+01	4.58E+01
F_6	Mean	6.37E+02	6.87E+02	6.26E+02	6.80E+02	6.42E+02	6.18E+02	6.09E+02	6.43E+02	6.87E+02	6.05E+02
	Std	4.99E+00	6.37E+00	5.16E+00	3.25E+00	9.56E+00	3.90E+00	3.46E+00	4.03E+00	9.85E+00	4.30E+00
F_7	Mean	1.98E+03	3.50E+03	1.76E+03	3.58E+03	1.80E+03	1.27E+03	1.96E+03	2.87E+03	3.33E+03	1.10E+03
	Std	1.38E+02	2.34E+02	3.02E+02	1.30E+02	1.50E+02	7.15E+01	9.24E+01	2.29E+02	1.62E+02	1.03E+02
F_8	Mean	1.46E+03	2.15E+03	1.35E+03	1.91E+03	1.45E+03	1.30E+03	1.21E+03	1.49E+03	1.99E+03	1.00E+03
	Std	5.20E+01	1.37E+02	5.90E+01	4.69E+01	1.08E+02	8.91E+01	5.72E+01	7.12E+01	9.01E+01	3.84E+01
F_9	Mean	2.80E+04	5.14E+04	2.88E+04	4.07E+04	1.43E+04	1.89E+04	1.88E+04	4.09E+04	4.52E+04	2.04E+03
	Std	1.17E+04	7.50E+03	1.95E+04	4.99E+03	6.24E+03	7.98E+03	7.43E+03	6.81E+03	1.17E+04	6.09E+02
F_{10}	Mean	1.57E+04	2.51E+04	1.46E+04	1.90E+04	1.55E+04	1.46E+04	1.54E+04	1.91E+04	2.27E+04	1.40E+04
	Std	3.33E+03	2.13E+03	1.52E+03	1.35E+03	1.10E+03	1.47E+03	3.32E+03	8.22E+03	2.35E+03	1.07E+03
F_{11}	Mean	5.57E+04	3.03E+04	1.58E+04	3.43E+03	2.24E+03	3.14E+03	1.17E+05	9.69E+03	4.95E+04	1.30E+03
	Std	1.41E+04	3.60E+04	1.70E+04	2.89E+02	3.24E+02	3.18E+02	4.37E+04	3.46E+03	3.07E+04	7.74E+01
F_{12}	Mean	9.76E+09	2.39E+09	1.52E+10	3.76E+08	4.30E+06	2.08E+08	1.24E+07	1.05E+10	1.16E+09	1.24E+05
	Std	3.69E+09	1.03E+09	1.06E+10	1.16E+08	1.80E+06	8.52E+07	7.27E+06	3.40E+09	4.77E+08	5.43E+04
F_{13}	Mean	1.16E+09	5.45E+07	1.17E+09	5.89E+06	1.51E+04	1.32E+05	1.02E+04	2.78E+08	1.08E+06	1.15E+04
	Std	1.19E+09	3.07E+07	1.00E+09	1.12E+06	9.63E+03	9.12E+04	6.63E+03	3.07E+08	8.45E+05	1.16E+04
F_{14}	Mean	6.08E+06	8.05E+06	2.16E+06	1.27E+06	4.49E+05	1.27E+06	1.76E+05	7.14E+05	3.53E+06	1.10E+04
	Std	3.53E+06	1.32E+07	1.32E+06	4.09E+05	2.78E+05	5.60E+05	1.16E+05	6.17E+05	1.10E+06	5.18E+03
F_{15}	Mean	1.13E+08	1.49E+07	3.33E+08	1.64E+06	7.91E+03	3.74E+04	4.51E+03	3.35E+06	3.24E+05	7.37E+03
	Std	1.30E+08	3.52E+07	5.72E+08	5.36E+05	5.97E+03	1.77E+04	4.44E+03	8.45E+06	6.22E+05	8.01E+03
F_{16}	Mean	5.91E+03	9.85E+03	6.07E+03	7.33E+03	5.61E+03	6.09E+03	5.19E+03	5.76E+03	1.18E+04	3.52E+03
	Std	5.45E+02	1.52E+03	8.78E+02	7.58E+02	6.42E+02	6.90E+02	6.80E+02	6.81E+02	1.81E+03	6.46E+02
F_{17}	Mean	4.75E+03	7.08E+03	6.54E+03	6.35E+03	4.99E+03	5.20E+03	4.68E+03	5.12E+03	7.64E+03	3.57E+03
	Std	6.70E+02	1.35E+03	1.26E+03	5.04E+02	6.09E+02	6.21E+02	5.56E+02	8.53E+02	1.23E+03	6.13E+02
F_{18}	Mean	4.73E+06	8.40E+06	3.11E+06	2.88E+06	1.24E+06	2.32E+06	1.56E+06	1.38E+06	3.25E+06	3.25E+04
	Std	2.86E+06	8.41E+06	1.43E+06	9.95E+05	7.10E+05	1.24E+06	1.83E+06	7.14E+05	1.35E+06	1.92E+04
F_{19}	Mean	2.97E+08	1.54E+07	3.72E+08	5.52E+06	9.77E+03	3.21E+04	5.88E+03	7.03E+06	2.24E+07	9.38E+03
	Std	3.82E+08	1.53E+07	4.72E+08	2.37E+06	9.25E+03	1.66E+04	3.32E+03	8.61E+06	1.93E+07	8.10E+03
F_{20}	Mean	4.59E+03	5.89E+03	4.85E+03	5.72E+03	4.92E+03	4.96E+03	4.80E+03	5.00E+03	6.21E+03	3.95E+03
	Std	7.91E+02	6.11E+02	6.56E+02	6.36E+02	6.11E+02	4.45E+02	8.87E+02	1.32E+03	5.39E+02	7.40E+02
F_{21}	Mean	2.95E+03	3.91E+03	3.11E+03	3.90E+03	2.98E+03	2.80E+03	2.78E+03	3.07E+03	3.96E+03	2.54E+03
	Std	5.70E+01	1.85E+02	1.45E+02	2.12E+02	1.08E+02	8.65E+01	7.32E+01	9.90E+01	2.49E+02	4.46E+01
F_{22}	Mean	1.87E+04	2.73E+04	1.73E+04	2.30E+04	1.76E+04	1.77E+04	1.69E+04	1.84E+04	2.58E+04	1.58E+04
	Std	3.20E+03	2.35E+03	1.36E+03	1.02E+03	1.73E+03	1.54E+03	1.84E+03	6.17E+03	2.87E+03	1.66E+03
F_{23}	Mean	3.56E+03	5.08E+03	4.62E+03	4.76E+03	3.36E+03	3.30E+03	3.24E+03	3.88E+03	4.87E+03	3.05E+03
	Std	8.35E+01	3.55E+02	2.53E+02	2.03E+02	8.28E+01	7.44E+01	6.89E+01	1.27E+02	2.62E+02	4.97E+01
F_{24}	Mean	4.20E+03	6.64E+03	5.90E+03	5.76E+03	3.91E+03	3.78E+03	3.73E+03	4.95E+03	6.03E+03	3.58E+03
	Std	1.49E+02	5.63E+02	4.41E+02	2.85E+02	1.10E+02	7.79E+01	1.04E+02	2.90E+02	3.37E+02	8.00E+01
F_{25}	Mean	6.55E+03	4.98E+03	5.01E+03	3.55E+03	3.32E+03	3.39E+03	3.32E+03	9.50E+03	3.99E+03	3.26E+03
	Std	9.57E+02	5.42E+02	1.27E+03	8.14E+01	6.71E+01	6.72E+01	7.72E+01	1.04E+03	1.45E+02	6.04E+01
F_{26}	Mean	1.47E+04	3.21E+04	2.07E+04	2.28E+04	1.24E+04	1.13E+04	1.02E+04	2.15E+04	3.28E+04	9.06E+03
	Std	1.11E+03	5.02E+03	4.53E+03	5.94E+03	1.31E+03	8.92E+02	8.56E+02	2.43E+03	3.42E+03	8.52E+02
F_{27}	Mean	4.04E+03	4.70E+03	4.24E+03	4.14E+03	3.61E+03	3.61E+03	3.45E+03	4.59E+03	5.05E+03	3.47E+03
	Std	1.72E+02	5.11E+02	3.60E+02	2.88E+02	8.05E+01	7.17E+01	4.71E+01	2.03E+02	6.57E+02	6.72E+01
F_{28}	Mean	8.16E+03	5.66E+03	1.02E+04	3.59E+03	3.39E+03	3.49E+03	3.42E+03	1.30E+04	4.41E+03	3.29E+03
	Std	1.06E+03	9.82E+02	3.50E+03	3.74E+01	3.32E+01	3.69E+01	2.94E+01	1.79E+03	1.99E+02	1.03E+02
F_{29}	Mean	8.17E+03	1.28E+04	7.57E+03	9.18E+03	6.56E+03	7.18E+03	6.27E+03	8.92E+03	1.47E+04	5.06E+03
	Std	7.43E+02	1.42E+03	1.37E+03	8.47E+02	5.47E+02	5.37E+02	4.27E+02	9.53E+02	1.62E+03	5.43E+02
F_{30}	Mean	1.13E+09	1.78E+08	1.45E+09	3.08E+07	1.33E+04	4.38E+06	2.17E+04	2.56E+08	3.85E+08	9.81E+03
	Std	1.36E+09	1.83E+08	1.39E+09	8.43E+06	5.43E+03	1.79E+06	1.80E+04	1.76E+08	1.63E+08	3.67E+03
Rank First		0	0	0	0	0	0	4	0	0	25

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